Chapter 9

Quadratic Programming

9.1 Optimality Conditions for Quadratic Programming

Quadratic programming is the simplest constrained nonlinear optimization problem. It is a special class of optimization problem (8.1.1)–(8.1.3) with a quadratic objective function $f(x)$ and linear constraints $c_i(x)$ $(i = 1, \cdots, m)$. The general quadratic programming (QP) has the following form:

$$\begin{aligned} \min \quad & Q(x) = \frac{1}{2}x^T Gx + g^T x && (9.1.1) \\ \text{s.t.} \quad & a_i^T x = b_i,\ i \in E, && (9.1.2) \\ & a_i^T x \geq b_i,\ i \in I, && (9.1.3) \end{aligned}$$

where G is a symmetric $n \times n$ matrix, E and I are finite sets of indices, $E = \{1, \cdots, m_e\}$ and $I = \{m_e+1, \cdots, m\}$. If the Hessian matrix G is positive semidefinite, then (9.1.1)–(9.1.3) is a convex quadratic programming problem and the local solution x^* is a global solution. If G is positive definite, then (9.1.1)–(9.1.3) is a strict convex QP and x^* is a unique global solution. If G is indefinite, then (9.1.1)–(9.1.3) is a nonconvex QP which is more important and worth emphasizing.

From Theorem 8.2.7, Theorem 8.3.3 and Theorem 8.3.4, we immediately get the following theorems:

Theorem 9.1.1 *(Necessary conditions)*

Let x^* be a local minimizer of quadratic programming problem (9.1.1)–(9.1.3). Then there exist multipliers λ_i^* $(i = 1, \cdots, m)$ such that

$$g + Gx^* = \sum_{i=1}^{m} \lambda_i^* a_i, \tag{9.1.4}$$

$$a_i^T x^* = b_i, \ i \in E, \tag{9.1.5}$$

$$a_i^T x^* \geq b_i, \ i \in I, \tag{9.1.6}$$

$$\lambda_i^*(a_i^T x^* - b_i) = 0, \ i \in I, \tag{9.1.7}$$

$$\lambda_i^* \geq 0, \ i \in I. \tag{9.1.8}$$

Furthermore,

$$d^T G d \geq 0, \ \forall d \in G(x^*, \lambda^*), \tag{9.1.9}$$

where

$$G(x^*, \lambda^*) = \left\{ d \neq 0 \;\middle|\; \begin{array}{l} d^T a_i = 0, \ i \in E \\ d^T a_i \geq 0, \ i \in I(x^*) \\ d^T a_i = 0, \ i \in I(x^*) \text{ and } \lambda_i^* > 0 \end{array} \right\}. \tag{9.1.10}$$

Theorem 9.1.2 *(Sufficient conditions)*

Let x^* be a KKT point and λ^* a corresponding Lagrange multiplier vector. If $d^T G d > 0$ $\forall\, 0 \neq d \in G(x^*, \lambda^*)$, then x^* is a strict local minimizer to (9.1.1)–(9.1.3).

Next, we give a sufficient and necessary optimality condition for (9.1.1)–(9.1.3).

Theorem 9.1.3 *(Necessary and sufficient conditions)*

Let x^* be a feasible point of quadratic programming problem (9.1.1)–(9.1.3), then x^* is a local minimizer if and only if (x^*, λ^*) is a KKT pair such that (9.1.4)–(9.1.8) hold, and

$$d^T G d \geq 0, \ \forall d \in G(x^*, \lambda^*). \tag{9.1.11}$$

Proof. Let x^* be a local minimizer, it follows from Theorem 9.1.1 that there exists multiplier vector λ^* such that (9.1.4)–(9.1.8) hold. Let $0 \neq d \in G(x^*, \lambda^*)$. Obviously, for sufficiently small $t > 0$, we have

$$x^* + td \in X. \tag{9.1.12}$$

Then, by the definition of d, for sufficiently small $t > 0$, we have

$$\begin{aligned} Q(x^*) &\leq Q(x^* + td) = Q(x^*) + td^T(Gx^* + g) + \frac{1}{2}t^2 d^T G d \\ &= Q(x^*) + t\sum_{i=1}^{m} \lambda_i^* a_i^T d + \frac{1}{2}t^2 d^T G d \\ &= Q(x^*) + \frac{1}{2}t^2 d^T G d, \end{aligned} \tag{9.1.13}$$

which, together with the arbitrariness of d, means (9.1.11) holds.

Second, we prove the sufficiency. Suppose, by contradiction, that x^* is not a local minimizer, so that there exists $x_k = x^* + \delta_k d_k \in X$ such that

$$Q(x_k) = Q(x^* + \delta_k d_k) < Q(x^*), \tag{9.1.14}$$

where $\delta_k > 0, \delta_k \to 0, d_k \to \bar{d}$. Completely similar to the proof of Theorem 8.3.4, we know that

$$\bar{d} \in G(x^*, \lambda^*). \tag{9.1.15}$$

Thus, it follows from (9.1.14) and KKT conditions that

$$\begin{aligned} \mathcal{L}(x^*, \lambda^*) &> \mathcal{L}(x_k, \lambda^*) \\ &= \mathcal{L}(x^*, \lambda^*) + \frac{1}{2}\delta_k^2 d_k^T G d_k + o(\delta_k^2). \end{aligned} \tag{9.1.16}$$

Dividing both sides by δ_k^2 and taking the limit, we obtain

$$\bar{d}^T G \bar{d} < 0. \tag{9.1.17}$$

Noting that $\bar{d} \in G(x^*, \lambda^*)$, it follows that (9.1.17) contradicts the assumption (9.1.11). Then we complete the proof. $\square$

Obviously, finding the KKT point of a quadratic programming problem is equivalent to finding $x^* \in R^n, \lambda^* \in R^m$ such that (9.1.4)–(9.1.8) hold.

9.2 Duality for Quadratic Programming

In this section we give more detailed discussion on the duality of convex quadratic programming, because in some classes of practical problems we can take advantage of the special structure of the dual to solve the problems more efficiently.

Assume that G is a positive definite matrix. From the results in §9.1, we have known that solving the quadratic programming problem (9.1.1)–(9.1.3) is equivalent to solving (9.1.4)–(9.1.8). Write

$$y = A\lambda - g, \tag{9.2.1}$$

and

$$t_i = a_i^T x - b_i, \ i \in I, \tag{9.2.2}$$

where $A = [a_1, \cdots, a_m] \in R^{n\times m}, \lambda = [\lambda_1, \cdots, \lambda_m]^T \in R^m$. Note that (9.1.4) is just $y = Gx$ and that (9.1.5)–(9.1.6) become

$$A^T x - b = (0, \cdots, 0, t_{m_e+1}, \cdots, t_m)^T,$$

then (9.1.4)–(9.1.8) can be written as

$$\begin{aligned}
&\begin{bmatrix} -b \\ G^{-1}y \end{bmatrix} = \begin{bmatrix} -A^T \\ I \end{bmatrix} x + (0, \cdots, 0, t_{m_e+1}, \cdots, t_m, 0, \cdots, 0)^T & (9.2.3)\\
&A\lambda - y = g, & (9.2.4)\\
&\lambda_i \geq 0, \ i \in I, & (9.2.5)\\
&t_i\lambda_i = 0, \ i \in I, & (9.2.6)\\
&t_i \geq 0, \ i \in I. & (9.2.7)
\end{aligned}$$

By KKT conditions, it follows that (9.2.3)–(9.2.7) are equivalent to solving the problem

$$\begin{aligned}
\max_{\lambda,y} \quad & b^T\lambda - \frac{1}{2}y^T G^{-1} y \stackrel{Def}{=} \bar{Q}(\lambda, y) & (9.2.8)\\
\text{s.t.} \quad & A\lambda - y = g, & (9.2.9)\\
& \lambda_i \geq 0, \ i \in I, & (9.2.10)
\end{aligned}$$

which is the dual of the primal (9.1.1)–(9.1.3). As an exercise, please prove that problem (9.2.8)–(9.2.10) just is

$$\begin{aligned}
\max_{x,\lambda} \quad & L(x, \lambda) = \frac{1}{2}x^T Gx + g^T x - \lambda^T(A^T x - b) & (9.2.11)\\
\text{s.t.} \quad & \nabla_x L(x, \lambda) = 0 & (9.2.12)\\
& \lambda_i \geq 0, \ i \in I. & (9.2.13)
\end{aligned}$$

Eliminating y in (9.2.9) by use of (9.2.1), we get that (9.2.8)–(9.2.10) can be reduced to

$$\min_{\lambda \in R^m} \quad -(b + A^T G^{-1} g)^T \lambda + \frac{1}{2}\lambda^T (A^T G^{-1} A)\lambda \tag{9.2.14}$$

$$\text{s.t.} \quad \lambda_i \geq 0, \ i \in I. \tag{9.2.15}$$

Assume that x and (λ, y) are feasible points of the primal problem (9.1.1)–(9.1.3) and the dual problem (9.2.8)–(9.2.10) respectively, then we have

$$\begin{aligned} Q(x) - \bar{Q}(\lambda, y) &= x^T(A\lambda - y) + \frac{1}{2}x^T G x \\ &\quad -(\lambda^T A x - \sum_{i \in I} \lambda_i t_i - \frac{1}{2} y^T G^{-1} y) \\ &= \sum_{i \in I} \lambda_i t_i + \frac{1}{2}(x^T G x + y^T G^{-1} y - 2x^T y), \end{aligned} \tag{9.2.16}$$

where t_i is defined in (9.2.2). Then, the positive definiteness of G gives

$$Q(x) \geq \bar{Q}(\lambda, y), \tag{9.2.17}$$

which is what we showed in Theorem 8.4.2.

It also follows from (9.2.16) that both sides of (9.2.17) are equal if and only if

$$\sum_{i \in I} \lambda_i (a_i^T x - b_i) = 0 \tag{9.2.18}$$

and

$$x = G^{-1} y. \tag{9.2.19}$$

It is not difficult to see that (9.2.19) and (9.2.18) are equivalent to (9.1.4) and (9.1.7) respectively. So, with the assumptions of feasibility, we have the following theorem.

Theorem 9.2.1 *Let G be positive definite. If the primal problem is feasible, then $x^* \in X$ is a solution of primal problem (9.1.1)–(9.1.3) if and only if (λ^*, y^*) is the solution of dual problem (9.2.8)–(9.2.10).*

In §8.4, we have shown that an unbounded primal implies an infeasible dual. We would like to know whether or not an infeasible primal implies an unbounded dual. This guess does not always hold. However, it is true for linearly constrained problems.

Theorem 9.2.2 *Let G be positive definite. Then the primal problem (9.1.1)–(9.1.3) is infeasible if and only if the dual (9.2.8)–(9.2.10) is unbounded.*

Proof. From (9.2.17), if the primal problem is feasible, the objective function of the dual problem on set satisfying constraints (9.2.9)–(9.2.10) is uniformly bounded above.

Now suppose that the primal problem is infeasible, then the system

$$(a_i^T, b_i)\tilde{x} = 0, \ i \in E, \tag{9.2.20}$$
$$(a_i^T, b_i)\tilde{x} \geq 0, \ i \in I, \tag{9.2.21}$$
$$(0, \cdots, 0, 1)\tilde{x} < 0 \tag{9.2.22}$$

has no solution for $\tilde{x} \in R^{n+1}$. By Corollary 8.2.6 of Farkas' Lemma, it follows that there exist $\bar{\lambda}_i$ $(i = 1, \cdots, m)$ such that

$$(0, \cdots, 0, 1) = \sum_{i \in E} \bar{\lambda}_i (a_i^T, b_i) + \sum_{i \in I} \bar{\lambda}_i (a_i^T, b_i), \tag{9.2.23}$$

i.e.,

$$\sum_{i=1}^{m} \bar{\lambda}_i a_i = 0, \tag{9.2.24}$$
$$\sum_{i=1}^{m} \bar{\lambda}_i b_i = 1, \tag{9.2.25}$$
$$\bar{\lambda}_i \geq 0, \ i \in I. \tag{9.2.26}$$

Set $\lambda_i = t\bar{\lambda}_i$, then (9.2.24) gives $A\lambda = tA\bar{\lambda} = 0$. It follows from (9.2.9) that $y = -g$. Therefore, when $t \to +\infty$, it follows from (9.2.8) and (9.2.25) that

$$\bar{Q}(\lambda, y) = t \to +\infty.$$

Also, for all $t > 0$, we have that $\lambda = (t\bar{\lambda}_1, \cdots, t\bar{\lambda}_m)^T$ and $y = -g$ satisfy constraints (9.2.9)–(9.2.10) of the dual problem. This shows that the dual problem is unbounded. $\Box$

There is a closed connection between Lagrangian function

$$\mathcal{L}(x, \lambda) = Q(x) - \sum_{i=1}^{m} \lambda_i (a_i^T x - b_i) \tag{9.2.27}$$

of the primal problem and duality. It is not difficult to see that solving KKT conditions is equivalent to finding a stationary point of $\mathcal{L}(x, \lambda)$ on the area $\{(x, \lambda) \,|\, \lambda_i \geq 0,\ i \in I\}$. Since the Hessian matrix of $\mathcal{L}(x, \lambda)$ is

$$\nabla^2 \mathcal{L}(x, \lambda) = \begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix}, \tag{9.2.28}$$

by use of the identity

$$\begin{bmatrix} I & 0 \\ A^T G^{-1} & I \end{bmatrix} \nabla^2 \mathcal{L}(x, \lambda) \begin{bmatrix} I & G^{-1}A \\ 0 & I \end{bmatrix} = \begin{bmatrix} G & 0 \\ 0 & -A^T G^{-1} A \end{bmatrix}, \tag{9.2.29}$$

we know that $\nabla^2 \mathcal{L}(x, \lambda)$ has just n positive eigenvalues, and that the number of negative eigenvalues equals rank(A). Thus, in general, the stationary point of $\mathcal{L}(x, \lambda)$ is a saddle point, i.e., there is $\lambda^* \in \Lambda$,

$$\Lambda = \{\lambda \in R^m \mid \lambda_i \geq 0,\ i \in I\},$$

such that (x^*, λ^*) satisfies

$$\mathcal{L}(x^*, \lambda) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) \tag{9.2.30}$$

for all $x \in X$ and $\lambda \in \Lambda$.

In fact, for all $x \in X$, we have

$$\max_{\lambda \in \Lambda} \mathcal{L}(x, \lambda) = Q(x). \tag{9.2.31}$$

For any $\lambda \in \Lambda$, set

$$y = A\lambda - g, \tag{9.2.32}$$

then (λ, y) is a feasible point of dual problem (9.2.8)–(9.2.10). This means that such a feasible (λ, y) satisfies (9.2.9), which is $\nabla_x \mathcal{L}(x, \lambda) = 0$, i.e., such a feasible (λ, y) such that $\min_{x \in R^n} \mathcal{L}(x, \lambda)$. Therefore,

$$\min_{x \in R^n} \mathcal{L}(x, \lambda) = b^T \lambda - \frac{1}{2} y^T G^{-1} y = \bar{Q}(\lambda, y). \tag{9.2.33}$$

Let (x^*, λ^*) be a solution of (9.1.4)–(9.1.8). Let $y^* = A\lambda^* - g$. It follows that (λ^*, y^*) is a feasible point of (9.2.8)–(9.2.10). Then, for any $x \in X$ and any $\lambda \in \Lambda$, we have

$$\begin{aligned} \mathcal{L}(x, \lambda^*) &\geq \bar{Q}(\lambda^*, y^*) \\ &= \mathcal{L}(x^*, \lambda^*) = Q(x^*) \geq \mathcal{L}(x^*, \lambda), \end{aligned} \tag{9.2.34}$$

which means that (x^*, λ^*) is a saddle point of $\mathcal{L}(x, \lambda)$.

Conversely, if

$$\mathcal{L}(x^*, \lambda) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) \tag{9.2.35}$$

holds for all $x \in X$ and $\lambda \in \Lambda$, then

$$\begin{aligned} Q(x^*) - \sum_{i=1}^m \lambda_i (a_i^T x^* - b_i) &\leq Q(x^*) - \sum_{i=1}^m \lambda_i^* (a_i^T x^* - b_i) \\ &\leq Q(x) - \sum_{i=1}^m \lambda_i^* (a_i^T x - b_i). \end{aligned} \tag{9.2.36}$$

Rearranging the first inequality gives

$$\sum_{i=1}^m (\lambda_i - \lambda_i^*)(a_i^T x^* - b_i) \geq 0, \tag{9.2.37}$$

which is

$$\sum_{i \in E} (\lambda_i - \lambda_i^*)(a_i^T x^* - b_i) + \sum_{i \in I} (\lambda_i - \lambda_i^*)(a_i^T x^* - b_i) \geq 0. \tag{9.2.38}$$

Now we prove

$$a_i^T x^* = b_i, \qquad i \in E \tag{9.2.39}$$

$$a_i^T x^* \geq b_i, \qquad i \in I \tag{9.2.40}$$

by contradiction. Suppose that $a_k^T x^* > b_k$ for some $k \in E$. Set $\lambda_i = \lambda_i^*$ for $i \neq k$ and $\lambda_k = \lambda_k^* - 1$, then we get a contradiction from (9.2.38) to the assumption $a_k^T x^* > b_k$. If we suppose $a_k^T x^* < b_k$ for some $k \in E$, we can get a similar contradiction. Therefore, we have that $a_i^T x^* = b_i$, $\forall i \in E$.

Now assume, for some $k \in I$, that

$$\lambda_k = \lambda_k^* + 1 \text{ and } \lambda_i = \lambda_i^* \text{ for } i \neq k. \tag{9.2.41}$$

Obviously, it follows that

$$a_k^T x^* - b_k \geq 0, \ k \in I. \tag{9.2.42}$$

Repeating the process for all $k \in I$, we obtain

$$a_i^T x^* - b_i \geq 0, \ \forall i \in I. \tag{9.2.43}$$

Then, x^* is a feasible point to the primal problem.

Set $\lambda = 0$, it follows from (9.2.35) that $\mathcal{L}(x^*, \lambda^*) \geq \mathcal{L}(x^*, 0)$, which is

$$\sum_{i=1}^{m} \lambda_i^* (a_i^T x^* - b_i) \leq 0. \tag{9.2.44}$$

By use of (9.2.44), (9.2.35) and $\lambda^* \in \Lambda$, we have, for all $x \in X$,

$$\begin{aligned} Q(x^*) &\leq Q(x^*) - \sum_{i=1}^{m} \lambda_i^* (a_i^T x^* - b_i) \\ &= \mathcal{L}(x^*, \lambda^*) \\ &\leq \mathcal{L}(x, \lambda^*) \\ &\leq \mathcal{L}(x, \lambda^*) + \sum_{i=1}^{m} \lambda_i^* (a_i^T x - b_i) \\ &= Q(x), \end{aligned} \tag{9.2.45}$$

which shows that x^* is a minimizer of the primal problem.

Therefore, we get the following theorem which is a famous saddle point theorem on the relationship between the saddle point of a Lagrangian function and the minimizer of the primal problem.

Theorem 9.2.3 *(Saddle point theorem for quadratic programming)*

Let G be positive definite. Then $x^* \in X$ is a minimizer of the primal problem (9.1.1)–(9.1.3) if and only if there exists $\lambda^* \in \Lambda$ such that (x^*, λ^*) is a saddle point of Lagrangian function $\mathcal{L}(x, \lambda)$, i.e., the saddle point conditions

$$\mathcal{L}(x^*, \lambda) \leq \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) \tag{9.2.46}$$

hold for all $x \in X$ and $\lambda \in \Lambda$.

9.3 Equality-Constrained Quadratic Programming

The equality-constrained quadratic programming problem can be written as

$$\min_{x \in R^n} \quad Q(x) = g^T x + \frac{1}{2} x^T G x \tag{9.3.1}$$

$$\text{s.t.} \quad A^T x = b, \tag{9.3.2}$$

where $g \in R^n$, $b \in R^m$, $A = [a_1, \cdots, a_m] \in R^{n\times m}, G \in R^{n\times n}$ and G is symmetric. Without loss of generality, we assume that rank $(A) = m$, i.e., A has full column rank.

First, we introduce the variable elimination method. Assume that the partitions are as follows:

$$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix}, \ A = \begin{bmatrix} A_B \\ A_N \end{bmatrix}, \ g = \begin{pmatrix} g_B \\ g_N \end{pmatrix}, \ G = \begin{bmatrix} G_{BB} & G_{BN} \\ G_{NB} & G_{NN} \end{bmatrix}, \tag{9.3.3}$$

where $x_B \in R^m, x_N \in R^{n-m}$, and A_B is invertible. By these partitions, the constraint condition (9.3.2) can be written as

$$A_B^T x_B + A_N^T x_N = b. \tag{9.3.4}$$

Since A_B^{-1} exists, then

$$x_B = (A_B^{-1})^T (b - A_N^T x_N). \tag{9.3.5}$$

Substituting it into (9.3.1) gives the following form

$$\min_{x_N \in R^{n-m}} \frac{1}{2} x_N^T \hat{G}_N x_N + \hat{g}_N^T x_N + \hat{c}, \tag{9.3.6}$$

which is equivalent to (9.3.1), where

$$\hat{g}_N = g_N - A_N A_B^{-1} g_B + [G_{NB} - A_N A_B^{-1} G_{BB}](A_B^{-1})^T b, \tag{9.3.7}$$

$$\begin{aligned} \hat{G}_N = {} & G_{NN} - G_{NB}(A_B^{-1})^T A_N^T \\ & - A_N A_B^{-1} G_{BN} + A_N A_B^{-1} G_{BB} (A_B^{-1})^T A_N^T, \end{aligned} \tag{9.3.8}$$

$$\hat{c} = \frac{1}{2} b^T A_B^{-1} G_{BB} A_B^{-T} b + g_B^T A_B^{-T} b. \tag{9.3.9}$$

If $\hat{G}_N$ is positive definite, the solution of (9.3.6) is

$$x_N^* = -\hat{G}_N^{-1} \hat{g}_N \tag{9.3.10}$$

which is unique. So the solution of problem (9.3.1)–(9.3.2) is

$$x^* = \begin{bmatrix} x_B^* \\ x_N^* \end{bmatrix} = \begin{bmatrix} (A_B^{-1})^T b \\ 0 \end{bmatrix} + \begin{bmatrix} (A_B^{-1})^T A_N^T \\ -I \end{bmatrix} \hat{G}_N^{-1} \hat{g}_N. \tag{9.3.11}$$

Let λ^* be the Lagrange multiplier vector at x^*, then

$$g + G x^* = A \lambda^*, \tag{9.3.12}$$

and thus

$$\lambda^* = A_B^{-1}(g_B + G_{BB}x_B^* + G_{BN}x_N^*). \tag{9.3.13}$$

If $\hat{G}_N$ in (9.3.6) is positive semi-definite, then when

$$(I - \hat{G}_N\hat{G}_N^+)\hat{g}_N = 0, \tag{9.3.14}$$

i.e., $\hat{g}_N \in R(\hat{G}_N)$, the minimization problem (9.3.6) is bounded, and its solution is

$$x_N^* = -\hat{G}_N^+\hat{g}_N + (I - \hat{G}_N^+\hat{G}_N)\tilde{x}, \tag{9.3.15}$$

where $\tilde{x} \in R^{n-m}$ is any vector, $\hat{G}_N^+$ denotes the generalized inverse matrix of $\hat{G}_N$. In this case, the solution of problem (9.3.1)–(9.3.2) can be represented by (9.3.15) and (9.3.5). If (9.3.14) does not hold, the problem (9.3.6) has no lower bound, and thus the original problem (9.3.1)–(9.3.4) also has no lower bound, that is, the original problem has no finite solution.

If $\hat{G}_N$ has negative eigenvalue, it is obvious that the minimization problem (9.3.6) has not lower bound, and thus the problem (9.3.1)–(9.3.2) has not finite solution.

Example 9.3.1

$$\begin{aligned} \min \quad & Q(x) = x_1^2 - x_2^2 - x_3^2 && (9.3.16)\\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1, && (9.3.17)\\ & x_2 - x_3 = 1. && (9.3.18) \end{aligned}$$

From (9.3.18), we have

$$x_2 = x_3 + 1. \tag{9.3.19}$$

Substituting it into (9.3.17) yields

$$x_1 = -2x_3. \tag{9.3.20}$$

In fact, here $x_B = (x_1, x_2)^T, x_N = x_3$. By substituting (9.3.19)–(9.3.20) into (9.3.16), we obtain

$$\min_{x_3 \in R} 4x_3^2 - (x_3 + 1)^2 - x_3^2. \tag{9.3.21}$$

Solving (9.3.21) gives $x_3 = \frac{1}{2}$. By substituting $x_3 = \frac{1}{2}$ into (9.3.19)–(9.3.20), we get

$$x^* = (-1, \frac{3}{2}, \frac{1}{2})^T,$$

which is the solution of (9.3.16)–(9.3.18).

By use of $g^* = A\lambda^*$, it follows that

$$\begin{pmatrix} -2 \\ -3 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \lambda_1^* \\ \lambda_2^* \end{pmatrix} \tag{9.3.22}$$

which gives Lagrange multipliers $\lambda_1^* = -2$ and $\lambda_2^* = -1$. $\square$

The idea of variable elimination method is simple and clear. However, when A_B closes to singular, computing the solution by (9.3.11) will lead to a numerically instable case.

A direct generalization of the variable elimination method is the generalized elimination method. We partition R^n into two complementary subspaces, i.e., $R^n = R(A) \oplus N(A^T)$. Let $y_1, \cdots, y_m$ be a set of linearly independent vectors in $R(A)$, the range of A, and let $z_1, \cdots, z_{n-m}$ be a set of linearly independent vectors in $N(A^T)$, the null space of A^T. Write

$$Y = [y_1, \cdots, y_m], \ Z = [z_1, \cdots, z_{n-m}],$$

which are $n \times m$ and $n \times (n-m)$ matrices respectively. Obviously, $R(Y) = R(A)$, $R(Z) = N(A^T)$, and $[Y : Z]$ is nonsingular. In addition, A^TY is nonsingular and $A^TZ = 0$. Set

$$x = Y\bar{x} + Z\hat{x}, \tag{9.3.23}$$

where $\bar{x} \in R^m, \hat{x} \in R^{n-m}$, it follows from the constraint condition (9.3.2) that

$$b = A^Tx = A^TY\bar{x}. \tag{9.3.24}$$

Then the feasible point of (9.3.1)–(9.3.2) can be represented as

$$x = Y(A^TY)^{-1}b + Z\hat{x}. \tag{9.3.25}$$

By substituting (9.3.25) into (9.3.1), we obtain

$$\min_{\hat{x} \in R^{n-m}} (g + GY(A^TY)^{-1}b)^T Z\hat{x} + \frac{1}{2}\hat{x}^T Z^T GZ\hat{x}, \tag{9.3.26}$$

which is an unconstrained minimization problem in R^{n-m}. Here Z^TGZ and $Z^T(g + GY(A^TY)^{-1}b)$ are called reduced Hessian and reduced gradient, respectively. Suppose that Z^TGZ is positive definite, then it follows from (9.3.26) that

$$(Z^TGZ)\hat{x} = -[Z^TGY(A^TY)^{-1}b + Z^Tg] \tag{9.3.27}$$

or

$$\hat{x}^* = -(Z^T G Z)^{-1} Z^T (g + GY(A^T Y)^{-1} b). \tag{9.3.28}$$

The system (9.3.27) can be solved by means of Cholesky factorization. Thus, the (9.3.28) and (9.3.25) give the solution of problem (9.3.1)–(9.3.2)

$$\begin{aligned} x^* &= Y(A^T Y)^{-1} b - Z(Z^T G Z)^{-1} Z^T (g + GY(A^T Y)^{-1} b) \\ &= (I - Z(Z^T G Z)^{-1} Z^T G) Y (A^T Y)^{-1} b - Z(Z^T G Z)^{-1} Z^T g. \end{aligned} \tag{9.3.29}$$

Furthermore, from the KKT condition

$$A\lambda^* = g + Gx^*,$$

by left-multiplying Y^T and noting that $A^T Y$ is nonsingular, we obtain

$$(Y^T A)\lambda^* = Y^T (g + Gx^*)$$

and

$$\begin{aligned} \lambda^* &= (A^T Y)^{-T} Y^T [g + Gx^*] \\ &= (A^T Y)^{-T} Y^T [Pg + GP^T Y (A^T Y)^{-1} b], \end{aligned} \tag{9.3.30}$$

where

$$P = I - GZ(Z^T G Z)^{-1} Z^T \tag{9.3.31}$$

is an affine mapping from R^n to $R(A)$. In particular, if we choose Y such that

$$A^T Y = I, \tag{9.3.32}$$

where Y is a left-inverse of A^T, then (9.3.25) becomes

$$x = Yb + Z\hat{x}, \tag{9.3.33}$$

where $\hat{x} \in R^{n-m}$, and further (9.3.29)–(9.3.30) become

$$x^* = Yb - Z(Z^T G Z)^{-1} Z^T (g + GYb) \tag{9.3.34}$$

$$= P^T Y b - Z(Z^T G Z)^{-1} Z^T g, \tag{9.3.35}$$

and

$$\begin{aligned} \lambda^* &= Y^T (g + Gx^*) \\ &= Y^T (Pg + GP^T Y b). \end{aligned} \tag{9.3.36}$$

From (9.3.25), we know that the feasible area of (9.3.1)–(9.3.2) is a subspace parallel to $N(A^T)$. The generalized elimination method just uses column-vectors z_i $(i = 1, \cdots, n-m)$ of Z, which form a base of the null space of A^T, as basis vectors and transforms the quadratic programming problem (9.3.1)–(9.3.2) into an unconstrained minimization (9.3.26) of quadratic function in a reduced space. Thus, this kind of method is also said to be null-space method.

The above discussions tell us that how to choose matrix Z, base matrix of the null space $N(A^T)$, is a key for this kind of methods. Different choices of Z form different null-space methods for solving quadratic programming problem (9.3.1)–(9.3.2). In the following we give some typical choices.

Clearly, the variable elimination method is a particular case of the generalized elimination method in which

$$Y = \begin{bmatrix} A_B^{-1} \\ 0 \end{bmatrix}, \tag{9.3.37}$$

$$Z = \begin{bmatrix} -A_B^{-T} A_N^T \\ I \end{bmatrix}. \tag{9.3.38}$$

Another particular case is based on QR decomposition of A. Let

$$A = Q \begin{bmatrix} R \\ 0 \end{bmatrix} = [Q_1 \; Q_2] \begin{bmatrix} R \\ 0 \end{bmatrix} = Q_1 R, \tag{9.3.39}$$

where Q is an $n \times n$ orthogonal matrix, R is an $m \times m$ nonsingular upper triangular matrix. Therefore, we have a choice

$$Y = (A^+)^T = Q_1 R^{-T}, \; Z = Q_2. \tag{9.3.40}$$

A general scheme for choosing Y and Z is as follows. For any Y and Z with $A^T Y = I$ and $A^T Z = 0$,

$$A^T [Y \; Z] = [I \; 0]. \tag{9.3.41}$$

Since $[Y \; Z]$ is nonsingular, there exists $V \in R^{n \times (n-m)}$ such that

$$[Y \; Z] = \begin{bmatrix} A^T \\ V^T \end{bmatrix}^{-1}, \tag{9.3.42}$$

i.e.,

$$[A\ V]^{-1} = \begin{bmatrix} Y^T \\ Z^T \end{bmatrix}. \tag{9.3.43}$$

It means that the different choices of $V \in R^{n\times(n-m)}$ lead to different Y and Z, and different elimination methods. For example, if we set

$$V = \begin{bmatrix} 0 \\ I_{n-m} \end{bmatrix},$$

we can get the variable elimination method (9.3.11). If we set

$$V = Q_2,$$

the above orthogonal decomposition choice (9.3.40) is obtained. Normally, null-space method is very useful, especially for small and medium-sized problems and when the computation of the null-space matrix Z and the factors of Z^TGZ is not very expensive.

The Lagrange method for solving equality-constrained quadratic programming is based on KKT conditions, which are

$$g + Gx = A\lambda, \tag{9.3.44}$$
$$A^Tx = b. \tag{9.3.45}$$

The above system can be written in the matrix form

$$\begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = - \begin{bmatrix} g \\ b \end{bmatrix}. \tag{9.3.46}$$

Here

$$\begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix} \tag{9.3.47}$$

is a KKT matrix for quadratic programming (9.3.1)–(9.3.2). It is not difficult to show that if A has full column-rank and Z^TGZ is positive definite, then KKT matrix (9.3.47) is nonsingular.

Theorem 9.3.2 *Let $A \in R^{n\times m}$ be a full column-rank matrix. Assume that the reduced Hessian Z^TGZ is positive definite. Then the KKT matrix (9.3.47) is nonsingular. Furthermore, there exists a unique KKT pair (x^*, λ^*) such that equation (9.3.46) is satisfied.*

Proof. The proof is by contradiction. Suppose that KKT matrix (9.3.47) is singular, then there exists nonzero vector $(p, v) \neq 0$ such that

$$\begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix} = 0, \tag{9.3.48}$$

where $p \in R^n$ and $v \in R^m$. Clearly, we have $A^T p = 0$. By left-multiplying $\begin{bmatrix} p \\ v \end{bmatrix}^T$ on both sides of (9.3.48), we obtain

$$0 = \begin{bmatrix} p \\ v \end{bmatrix}^T \begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix} \begin{bmatrix} p \\ v \end{bmatrix} = p^T G p.$$

Since $p \in N(A^T)$ and $Z = [z_1, \cdots, z_{n-m}]$ spans $N(A^T)$, we may denote $p = Zu$ for some $u \in R^{n-m}$ and have

$$0 = p^T G p = u^T Z G Z u.$$

The assumption that $Z^T G Z$ is positive definite gives $u = 0$ and then

$$p = Zu = 0. \tag{9.3.49}$$

So, it follows from (9.3.48) that $Av = 0$. Notice that A has full column-rank, then we obtain also $v = 0$ which together with (9.3.49) contradicts the fact $(p, v) \neq 0$. We complete the proof. $\Box$

Now let KKT matrix be nonsingular. Then there exist matrices $U \in R^{n\times n}, W \in R^{n\times m}$ and $T \in R^{m\times m}$ such that

$$\begin{bmatrix} G & -A \\ -A^T & 0 \end{bmatrix}^{-1} = \begin{bmatrix} U & W \\ W^T & T \end{bmatrix}, \tag{9.3.50}$$

and the unique solution of (9.3.46) is

$$x^* = -Ug - Wb, \tag{9.3.51}$$
$$\lambda^* = -W^T g - Tb. \tag{9.3.52}$$

As long as the KKT matrix (9.3.47) is nonsingular, then (9.3.50) is determined uniquely, so the stationary point of the Lagrangian function is determined uniquely by (9.3.51)–(9.3.52). However, since there are many expressions for U, W, and T, and we can derive a different computational schemes of formula (9.3.51)–(9.3.52).

If G is invertible and A has full column-rank, then $(A^TG^{-1}A)^{-1}$ exists. It is not difficult to show that the expressions of U, W, and T in (9.3.50) are

$$U = G^{-1} - G^{-1}A(A^TG^{-1}A)^{-1}A^TG^{-1}, \tag{9.3.53}$$
$$W = -G^{-1}A(A^TG^{-1}A)^{-1}, \tag{9.3.54}$$
$$T = -(A^TG^{-1}A)^{-1}. \tag{9.3.55}$$

Then it follows from (9.3.46) that the solution for quadratic programming with equality constraints is

$$x^* = -G^{-1}g + G^{-1}A(A^TG^{-1}A)^{-1}[A^TG^{-1}g + b], \tag{9.3.56}$$
$$\lambda^* = (A^TG^{-1}A)^{-1}[A^TG^{-1}g + b]. \tag{9.3.57}$$

As we said, if A has full column-rank and Z^TGZ is positive definite, then KKT matrix is invertible. In this case, if Y and Z are defined by (9.3.42), the matrices U, W, and T in (9.3.50) can be represented as

$$U = Z(Z^TGZ)^{-1}z^T, \tag{9.3.58}$$
$$W = -P^TY, \tag{9.3.59}$$
$$T = -Y^TGP^TY, \tag{9.3.60}$$

where P is defined by (9.3.31). Substituting (9.3.58)–(9.3.60) into (9.3.51)–(9.3.52) yields the formula (9.3.35)–(9.3.36). Hence, the Lagrange method is equivalent to the generalized elimination method.

9.4 Active Set Methods

Most QP problems involve inequality constraints and so can be expressed in the form (9.1.1)–(9.1.3). In this section we describe how the methods for solving equality-constrained QP can be generalized to handle the general QP problem (9.1.1)–(9.1.3) by means of active set methods, which are, in general, the most effective methods for small and medium-sized problems. We start our discussion by considering the convex case, i.e., the matrix G in (9.1.1)–(9.1.3) is positive semi-definite. The other case in which G is indefinite will be simply discussed in the end of the section. Intuitively, inactive inequality constraints do not play any role near the solution, so they can be dropped; the active inequality constraints have zero values at solution, and so they can be replaced by equality constraints. The following lemma is a base for active set methods.

Lemma 9.4.1 *Let x^* be a local minimizer of QP problem (9.1.1)–(9.1.3). Then x^* is a local minimizer of problem*

$$\min_{x\in R^n} \quad g^T x + \frac{1}{2}x^T Gx \tag{9.4.1}$$

$$\text{s.t.} \qquad a_i^T x = b_i,\ i \in E \cup I(x^*). \tag{9.4.2}$$

Conversely, if x^* is a feasible point of (9.1.1)–(9.1.3) and a KKT point of (9.4.1)–(9.4.2), and the corresponding Lagrange multiplier vector λ^* satisfies

$$\lambda_i^* \geq 0,\ i \in I(x^*), \tag{9.4.3}$$

then x^* is also the KKT point of problem (9.1.1)–(9.1.3).

Proof. Since, near x^*, the feasible point of (9.1.1)–(9.1.3) is also feasible for problem (9.4.1)–(9.4.2), then, obviously, the local minimizer of (9.1.1)–(9.1.3) is also the local minimizer of problem (9.4.1)–(9.4.2).

Now let x^* be feasible for (9.1.1)–(9.1.3) and a KKT point for (9.4.1)–(9.4.2). Let there exist λ_i^* ($i \in E \cup I(x^*)$) such that

$$Gx^* + g = \sum_{i\in I(x^*)\cup E} a_i\lambda_i^*, \tag{9.4.4}$$

$$\lambda_i^*(a_i^T x^* - b_i) = 0,\ \lambda_i^* \geq 0,\ i \in I(x^*). \tag{9.4.5}$$

Define

$$\lambda_i^* = 0,\ i \in I \setminus I(x^*). \tag{9.4.6}$$

Then we immediately have from (9.4.4)–(9.4.6) that

$$Gx^* + g = \sum_{i=1}^{m} \lambda_i^* a_i, \tag{9.4.7}$$

$$a_i^T x^* = b_i,\ i \in E, \tag{9.4.8}$$

$$a_i^T x^* \geq b_i,\ i \in I, \tag{9.4.9}$$

$$\lambda_i^* \geq 0,\ i \in I, \tag{9.4.10}$$

$$\lambda_i^*(a_i^T x^* - b_i) = 0,\ \forall i \tag{9.4.11}$$

which means that x^* is a KKT point of problem (9.1.1)–(9.1.3). □

The active set methods are a feasible point method, that is, all iterates remain feasible. In each iteration, we solve a quadratic programming subproblem with a subset of equality constraints. This subset is said to be a working set and is denoted by $\mathcal{S}_k \subset E \cup I(x^*)$.

If the solution of the equality-constrained QP subproblem on $\mathcal{S}_k$ is feasible for original problem (9.1.1)–(9.1.3), we need to examine whether (9.4.3) is satisfied or not. If (9.4.3) is satisfied, then stop and we get the solution of the original problem. Otherwise, the KKT conditions are not satisfied, and the objective function $q(\cdot)$ can be decreased by dropping this constraint. Thus, we remove the index from the working set $\mathcal{S}_k$ and solve a new subproblem. If the solution of equality-constrained QP subproblem on $\mathcal{S}_k$ is not feasible for problem (9.1.1)–(9.1.3), we need to add a constraint into the working set $\mathcal{S}_k$ and then solve a new subproblem.

At each iteration, a feasible point x_k and a working set $\mathcal{S}_k$ are known. Each iteration attempts to locate a solution of an equality-constrained subproblem on $\mathcal{S}_k$. Let d be a step from x_k. We can express the QP subproblem in terms of d. Consider the QP subproblem

$$\min_{d\in R^n} \quad \frac{1}{2}(x_k+d)^T G(x_k+d) + g^T(x_k+d), \tag{9.4.12}$$

$$\text{s.t.} \quad a_i^T d = 0,\ i \in \mathcal{S}_k \tag{9.4.13}$$

which can be written as

$$\min_{d\in R^n} \quad \frac{1}{2}d^T G d + g_k^T d \tag{9.4.14}$$

$$\text{s.t.} \quad a_i^T d = 0,\ i \in \mathcal{S}_k \tag{9.4.15}$$

where $g_k = \nabla Q(x_k) = Gx_k + g$. Denote the KKT point of (9.4.12)–(9.4.13) by d_k, the corresponding Lagrange multipliers by $\lambda_i^{(k)}$ $(i \in \mathcal{S}_k)$. If $d_k = 0$, then x_k is the KKT point of subproblem

$$\min_{x\in R^n} \quad \frac{1}{2}x^T G x + g^T x \tag{9.4.16}$$

$$\text{s.t.} \quad a_i^T x = b_i,\ i \in \mathcal{S}_k. \tag{9.4.17}$$

At this time, if $\lambda_i^{(k)} \geq 0, \forall i \in \mathcal{S}_k \cap I$, then x_k is a KKT point of problem (9.1.1)–(9.1.3), and we terminate the iteration. Otherwise, there exists negative Lagrange multiplier, for example, $\lambda_{i_k}^{(k)} < 0$. In this case, it is possible to reduce the objective function by dropping the i_k-th constraint from current working set $\mathcal{S}_k$. Then we solve the resulting QP subproblem. Note that if there are more than one index such that $\lambda_i < 0$, it is usual to choose i_k for which

$$\lambda_{i_k} = \min_{\substack{i\in\mathcal{S}_k\cap I\\ \lambda_i^{(k)}<0}} \lambda_i^{(k)} \tag{9.4.18}$$

and set

$$\mathcal{S}_k := \mathcal{S}_k \setminus \{i_k\}. \tag{9.4.19}$$

Suppose that the solution $d_k \neq 0$. If $x_k + d_k$ is feasible with regard to all the constraints, then we set

$$x_{k+1} = x_k + d_k. \tag{9.4.20}$$

Otherwise, a line search is made along the direction d_k and we set

$$x_{k+1} = x_k + \alpha_k d_k, \tag{9.4.21}$$

where α_k is a steplength such that $x_k + \alpha_k d_k$ is the "best" feasible point on $[x_k, x_k + d_k]$ and the closest to $x_k + d_k$, i.e., take α_k as large as possible in the interval $[0, 1]$.

Now we derive the explicit formula for α_k. We ask $x_k + \alpha_k d_k$ for satisfying all constraints. Obviously, if $i \in \mathcal{S}_k$, then the corresponding constraint will be certainly feasible. Thus we only need to consider those constraints for which $i \notin \mathcal{S}_k$. There are two cases we need to consider. If $a_i^T d_k \geq 0$ for some $i \notin \mathcal{S}_k$, then we have for all $\alpha_k \geq 0$,

$$a_i^T (x_k + \alpha_k d_k) \geq a_i^T x_k \geq b_i, \; i \notin \mathcal{S}_k.$$

In this case, the constraint is satisfied. If $a_i^T d_k < 0$ for some $i \notin \mathcal{S}_k$, we have

$$a_i^T (x_k + \alpha_k d_k) \geq b_i$$

only if

$$\alpha_k \leq \frac{b_i - a_i^T x_k}{a_i^T d_k}, \; i \notin \mathcal{S}_k. \tag{9.4.22}$$

Hence, we should take

$$\alpha_k = \min_{\substack{i \notin \mathcal{S}_k \\ a_i^T d_k < 0}} \frac{b_i - a_i^T x_k}{a_i^T d_k}. \tag{9.4.23}$$

Since we want α_k to be as large as possible in $[0, 1]$ subject to remaining feasibility, we have the following formula:

$$\alpha_k = \min \left\{ 1, \; \min_{\substack{i \notin \mathcal{S}_k \\ a_i^T d_k < 0}} \frac{b_i - a_i^T x_k}{a_i^T d_k} \right\}. \tag{9.4.24}$$

If $\alpha_k < 1$, i.e., (9.4.23) holds, then there exists some $j \notin \mathcal{S}_k$ such that

$$\alpha_k = \frac{b_j - a_j^T x_k}{a_j^T d_k}.$$

Thus,

$$a_j^T x_{k+1} = a_j^T x_k + \alpha_k a_j^T d_k = b_j.$$

This means that there is a new constraint indexed by $j \notin \mathcal{S}_k$ becoming an active constraint at x_{k+1}. So we put it into the working set, that is, set $\mathcal{S}_{k+1} = \mathcal{S}_k \cup \{j\}$.

If $\alpha_k = 1$, then the working set remains the same, i.e., $\mathcal{S}_{k+1} = \mathcal{S}_k$.

So, we can continue the next iteration on the new working set $\mathcal{S}_{k+1}$.

Now, we are in a position to give the algorithm of active set method as follows.

Algorithm 9.4.2 *(Active Set Methods)*

Step 1. Given x_1, set $\mathcal{S}_1 = E \cup I(x_1), k := 1$.

Step 2. Find the solution d_k for subproblem (9.4.12)–(9.4.13).
If $d_k \neq 0$, go to Step 3;
Else if $d_k = 0$, compute $\lambda_i^{(k)}$ from $Gx_k + g = \sum_{i \in \mathcal{S}_k} \lambda_i^{(k)} a_i$.
If $\lambda_i^{(k)} \geq 0 \, \forall i \in \mathcal{S}_k \cap I$, stop;
else find i_k by (9.4.18).
$\mathcal{S}_k := \mathcal{S}_k \setminus \{i_k\}, x_{k+1} = x_k$, go to Step 4.

Step 3. Find α_k by (9.4.24);
Set

$$x_{k+1} = x_k + \alpha_k d_k. \tag{9.4.25}$$

If $\alpha_k = 1$, go to Step 4;
Else find $j \notin \mathcal{S}_k$ such that

$$a_j^T (x_k + \alpha_k d_k) = b_j. \tag{9.4.26}$$

Set $\mathcal{S}_k := \mathcal{S}_k \cup \{j\}$.

Step 4. $\mathcal{S}_{k+1} := \mathcal{S}_k, k := k + 1$, go to Step 2. $\square$

Now we give analysis to the algorithm.

From Algorithm 9.4.2, we know that all iterates are feasible, i.e.,

$$x_k \in X, \ \forall k, \tag{9.4.27}$$

and the objective function remains descent, i.e.,

$$Q(x_{k+1}) \leq Q(x_k), \ \forall k. \tag{9.4.28}$$

Further, as long as $d_k \neq 0$ (i.e., x_k is not the KKT point of (9.4.16)–(9.4.17)) and $\alpha_k > 0$, we have

$$Q(x_{k+1}) < Q(x_k). \tag{9.4.29}$$

If the algorithm terminates in finitely many steps, the obtained point is a KKT point of the original problem (9.1.1)–(9.1.3).

Suppose that the algorithm does not terminate in finitely many steps; since there is only a finite number of constraints, it is impossible that the number of elements in $\mathcal{S}_k$ increases infinitely many times and does not reduce. So there are infinitely many indices k such that $d_k = 0$. It follows from the algorithm that there are infinitely many indices k such that x_k is a KKT point of (9.4.16)–(9.4.17). Since the number of constraints is finite, $\mathcal{S}_k$ has only finitely many different combinations and so the sequence of the objective values $\{Q(x_k)\}$ has only finitely many elements. Therefore, there must exist a sufficiently large k_0 such that

$$Q(x_{k+1}) = Q(x_k), \ \forall k \geq k_0. \tag{9.4.30}$$

Then for all $k \geq k_0$, in both

$$\alpha_k = 0 \tag{9.4.31}$$

and

$$d_k = 0, \tag{9.4.32}$$

only one holds. Since there are only finitely many constraints, it is impossible that the algorithm only increases the constraint into $\mathcal{S}_k$, nor reduces the constraint from $\mathcal{S}_k$. Hence, there must be infinitely many indices k such that

$$d_k \neq 0, \tag{9.4.33}$$

and infinitely many indices k such that

$$d_k = 0. \tag{9.4.34}$$

So, there exist $k_2 > k_1 > k_0$ such that

$$d_{k_1} = 0,\ d_{k_2} = 0, \tag{9.4.35}$$

$$d_k \neq 0,\ k_1 < k < k_2, \tag{9.4.36}$$

and

$$k_2 > k_1 + 1. \tag{9.4.37}$$

Lemma 9.4.3 *Let k_0 be an index satisfying (9.4.30). If $k_2 > k_1 > k_0$ satisfy (9.4.35)–(9.4.37), then*

$$\mathcal{S}_{k_2} \neq \mathcal{S}_{k_1}. \tag{9.4.38}$$

Proof. By (9.4.35), there exist $\lambda_i^{(k_1)}$ such that

$$g + G\bar{x} = \sum_{i \in \mathcal{S}_{k_1}} a_i \lambda_i^{(k_1)}, \tag{9.4.39}$$

where $\bar{x} = x_{k_0}$. From (9.4.31)–(9.4.32), it follows that $x_k = \bar{x}$ for all $k \geq k_0$.

Since $d_{k_1+1} \neq 0, \alpha_{k_1+1} = 0$, there must be

$$j \notin \mathcal{S}_{k_1+1}, \tag{9.4.40}$$

such that $j \in \mathcal{S}_{k_1+2}$,

$$j \in I(\bar{x}) \tag{9.4.41}$$

and

$$a_j^T d_{k_1+1} < 0. \tag{9.4.42}$$

Since d_k is a solution for subproblem (9.4.12)–(9.4.13), i.e., d_k is a descent direction of the objective function, then

$$(g + G\bar{x})^T d_{k_1+1} \leq 0. \tag{9.4.43}$$

By using (9.4.39), (9.4.43) and $\mathcal{S}_{k_1+1} = \mathcal{S}_{k_1} \setminus \{i_{k_1}\}$, we get

$$\lambda_{i_{k_1}}^{(k_1)} a_{i_{k_1}}^T d_{k_1+1} \leq 0, \tag{9.4.44}$$

which means that

$$a_{i_{k_1}}^T d_{k_1+1} \geq 0 \tag{9.4.45}$$

by the definition of $\{i_k\}$. Comparing (9.4.42)–(9.4.44) gives $j \neq i_{k_1}$. Hence it follows from (9.4.40) that $j \notin \mathcal{S}_{k_1}$.

On the other hand, $j \in \mathcal{S}_{k_1+2} \subseteq \mathcal{S}_{k_2}$. Hence we have $\mathcal{S}_{k_2} \neq \mathcal{S}_{k_1}$. The proof is complete. $\square$

Finally, we give the convergence theorem of active set methods.

Theorem 9.4.4 *If, for all k, $a_i\,(i \in E \cup I(x_k))$ are linearly independent, then either the sequence generated from Algorithm 9.4.2 converges to a KKT point of problem (9.1.1)–(9.1.3) in finite iterations, or the original problem (9.1.1)–(9.1.3) is unbounded below.*

Proof. Assume that the problem (9.1.1)–(9.1.3) is bounded below, then the sequence $\{x_k\}$ is bounded.

If the solution of subproblem (9.4.12)–(9.4.13) is $d_k = 0$, then x_k is a KKT point of (9.4.16)–(9.4.17) for the current working set $\mathcal{S}_k$. If $\lambda_i^{(k)} \geq 0$, $\forall i \in \mathcal{S}_k \cap I$, then x_k is a KKT point of the original problem (9.1.1)–(9.1.3). Otherwise, there exists $\lambda_{i_k}^{(k)} < 0$ $(i_k \in \mathcal{S}_k \cap I)$ for which we can find a feasible descent direction d_k such that

$$a_j^T d_k = 0, \quad j \in \mathcal{S}_k, j \neq i_k, \tag{9.4.46}$$

$$a_{i_k}^T d_k > 0 \tag{9.4.47}$$

and

$$g_k^T d_k = (\lambda^{(k)})^T A_k^T d_k = (a_{i_k}^T d_k)(\lambda^{(k)})^T e_{i_k} = (a_{i_k}^T d_k)\lambda_{i_k}^{(k)} < 0. \tag{9.4.48}$$

If we substitute (9.4.46) for the constraints in (9.4.13), i.e., set $\mathcal{S}_k := \mathcal{S}_k \backslash \{i_k\}$, the resulting QP subproblem will have a feasible descent direction. Since $\alpha_k > 0$, we have

$$Q(x_{k+1}) < Q(x_k),$$

and consequently, by finiteness of constraints, the algorithm never returns to the current working set $\mathcal{S}_k$, and the sequence $\{x_k\}$ is finite.

If $d_k \neq 0$ and $\alpha_k = 1$, then $\mathcal{S}_{k+1} = \mathcal{S}_k$, and the subproblem (9.4.12)–(9.4.13) is unchanged for x_{k+1} and so the x_{k+1} is the solution of (9.4.12)–(9.4.13).

Only if $d_k \neq 0$ and $\alpha_k < 1$, x_{k+1} is not the solution of (9.4.12)–(9.4.13). At this time, from (9.4.26) in Step 3 of Algorithm 9.4.2, we know that there is an index $j \notin \mathcal{S}_k$ such that the j-th constraint is feasible. So, such a constraint is added into $\mathcal{S}_{k+1}$. If this procedure occurs repeatedly, then after at most n iterations the working set $\mathcal{S}_k$ will contains n indices, which correspond to n linearly independent vectors, then it follows from (9.4.13) that $d_k = 0$. Thus such a procedure continues at most n times. So there will be a KKT point x_k of (9.4.16)–(9.4.17) at most after n iterations.

Combining the above discussion, in any case, the algorithm will converge in finite iterations to the KKT point of problem (9.1.1)–(9.1.3). $\square$

By modifying the algorithm, the active set method for a convex QP problem can be adopted to the indefinite case in which the Hessian matrix G has some negative eigenvalues.

As we know from §9.3 that if G in $\mathcal{S}_k$ is indefinite, then the problem (9.4.13) may be unbounded. We can choose the direction d_k such that $a_i^T d_k = 0\,(\forall i \in \mathcal{S}_k)$ and either

$$d_k^T G d_k < 0 \tag{9.4.49}$$

or

$$\nabla Q(x_k)^T d_k < 0,\ d_k^T G d_k = 0 \tag{9.4.50}$$

where $\nabla Q(x_k) = g + Gx_k$. If, for all $i \notin \mathcal{S}_k$, $a_i^T d_k \geq 0$, then the original problem (9.1.1)–(9.1.3) is unbounded below. Otherwise, we can find $i \notin \mathcal{S}_k$ and $a_i^T d_k < 0$. Then, when $\alpha > 0$ is sufficiently large, $x_k + \alpha d_k$ is not a feasible point of (9.1.1)–(9.1.3). In this case we can take α_k as large as possible and make $x_k + \alpha_k d_k$ feasible.

9.5 Dual Method

For the convex QP problem

$$\min_{x\in R^n} \quad g^T x + \frac{1}{2}x^T G x \tag{9.5.1}$$

$$\text{s.t.} \quad a_i^T x = b_i,\ i \in E, \tag{9.5.2}$$

$$a_i^T x \geq b_i,\ i \in I, \tag{9.5.3}$$

where G is symmetric and positive definite. We know from §9.2 that the dual problem is

$$\min_{\lambda\in R^m} \quad -(b + AG^{-1}g)^T\lambda + \frac{1}{2}\lambda^T(A^T G^{-1} A)\lambda \tag{9.5.4}$$

$$\text{s.t.} \quad \lambda_i \geq 0,\ i \in I. \tag{9.5.5}$$

Now we adopt the active-set method to (9.5.4)–(9.5.5). The equality-constrained subproblem we solved at each iteration is

$$\min_{\lambda\in R^m} \quad -(b + A^T G^{-1}g)^T\lambda + \frac{1}{2}\lambda^T(A^T G^{-1} A)\lambda \tag{9.5.6}$$

$$\text{s.t.} \quad \lambda_i = 0,\ i \in \bar{\mathcal{S}}_k, \tag{9.5.7}$$

where $\bar{\mathcal{S}}_k \subseteq I$ is a working set for dual problem (9.5.4)–(9.5.5). Let λ_k be a KKT point of the subproblem (9.5.6)–(9.5.7). Set

$$x_k = -G^{-1}(g - A\lambda_k), \tag{9.5.8}$$

then

$$Gx_k + g = A\lambda_k, \tag{9.5.9}$$

and from

$$(b + A^T G^{-1} g - A^T G^{-1} A\lambda_k)_i = 0, \ \forall i \notin \bar{\mathcal{S}}_k, \tag{9.5.10}$$

we have

$$(A^T x_k - b)_i = 0, \ \forall i \notin \bar{\mathcal{S}}_k. \tag{9.5.11}$$

Thus, x_k is the KKT point of the subproblem

$$\min_{x\in R^n} \quad g^T x + \frac{1}{2} x^T G x \tag{9.5.12}$$

$$\text{s.t.} \qquad a_i^T x = b_i, \ i \notin \bar{\mathcal{S}}_k. \tag{9.5.13}$$

Write $\mathcal{S}_k = \{I \cup E\} \setminus \bar{\mathcal{S}}_k$. It is obvious that (9.5.12)–(9.5.13) is the same as (9.4.16)–(9.4.17). It is not difficult to see that the Lagrange multipliers of dual problem (9.5.6)-(9.5.7) satisfy

$$\begin{aligned} & (A^T G^{-1} A\lambda_k - b - A^T G^{-1} g)_i \\ = \ & (A^T x_k - b)_i = a_i^T x_k - b_i, \ i \in \bar{\mathcal{S}}_k. \end{aligned} \tag{9.5.14}$$

We ask λ_k to be a feasible point of (9.5.4)–(9.5.5). If the Lagrange multiplier (9.5.14) of the dual problem (9.5.6)–(9.5.7) is nonnegative, x_k is a KKT point of the original problem (9.5.1)–(9.5.3). Let A_k be a matrix with the columns $a_i \, (i \in \mathcal{S}_k), \bar{\lambda}_k$ the vector consisting of the components of λ_k corresponding to $i \in \mathcal{S}_k$. It follows from (9.5.10) that

$$b_i + a_i^T G^{-1} g - a_i^T G^{-1} A_k \bar{\lambda}_k = 0, \ i \in \mathcal{S}_k, \tag{9.5.15}$$

i.e.,

$$b^{(k)} + A_k^T G^{-1} g - A_k^T G^{-1} A_k \bar{\lambda}_k = 0, \tag{9.5.16}$$

where $b^{(k)}$ consists of the components of b corresponding to $i \in \mathcal{S}_k$. Then (9.5.16) gives

$$\bar{\lambda}_k = (A_k^T G^{-1} A_k)^{-1} [b^{(k)} + A_k^T G^{-1} g]. \tag{9.5.17}$$

When Lagrange multipliers in (9.5.10) are not all nonnegative, we should, by the active-set method, drop an index $i_k \in \bar{\mathcal{S}}_k$, that is, add the index i_k into $\mathcal{S}_k$. For convenience of sign, we write i_k as p. Then we have $\mathcal{S}_{k+1} = \mathcal{S}_k \cup \{p\}$.

Let

$$\bar{\lambda}_{k+1} = \begin{pmatrix} \bar{\lambda}_k \\ 0 \end{pmatrix} + \begin{pmatrix} \delta\lambda_k \\ \beta_k \end{pmatrix}. \tag{9.5.18}$$

It follows from (9.5.17) that

$$\begin{pmatrix} A_k^T G^{-1} A_k & A_k^T G^{-1} a_p \\ a_p^T G^{-1} A_k & a_p^T G^{-1} a_p \end{pmatrix} \begin{pmatrix} \delta\lambda_k \\ \beta_k \end{pmatrix} = \begin{pmatrix} 0 \\ b_p - a_p^T x_k \end{pmatrix}, \tag{9.5.19}$$

which gives

$$\bar{\lambda}_{k+1} = \begin{pmatrix} \bar{\lambda}_k \\ 0 \end{pmatrix} + \beta_k \begin{pmatrix} -(A_k^T G^{-1} A_k)^{-1} A_k^T G^{-1} a_p \\ 1 \end{pmatrix}. \tag{9.5.20}$$

So,

$$\begin{aligned} x_{k+1} &= x_k + G^{-1} A_{k+1} \left(\bar{\lambda}_{k+1} - \begin{bmatrix} \bar{\lambda}_k \\ 0 \end{bmatrix} \right) \\ &= x_k + \beta_k G^{-1} (I - A_k (A_k^T G^{-1} A_k)^{-1} A_k^T G^{-1}) a_p. \end{aligned} \tag{9.5.21}$$

Let

$$A_k^* = (A_k^T G^{-1} A_k)^{-1} A_k^T G^{-1}, \tag{9.5.22}$$

$$y_k = A_k^* a_p. \tag{9.5.23}$$

Since $\bar{\lambda}_{k+1}$ should satisfy $\bar{\lambda}_{k+1} \geq 0$, it follows from (9.5.20) and (9.5.23) that

$$0 \leq \beta_k \leq \min_{\substack{j \in \mathcal{S}_k \\ (y_k)_j > 0}} \frac{(\bar{\lambda}_k)_j}{(y_k)_j}. \tag{9.5.24}$$

If

$$G^{-1}(I - A_k A_k^*) a_p = 0 \tag{9.5.25}$$

and $y_k \leq 0$, then

$$(-y_k, 1)^T (A_{k+1}^T G^{-1} A_{k+1}) \begin{pmatrix} -y_k \\ 1 \end{pmatrix} = 0 \tag{9.5.26}$$

and

$$(-y_k, 1)^T (b^{(k+1)} + A_{k+1}^T G^{-1} g) = b_p - a_p^T x_k > 0, \tag{9.5.27}$$

which indicate that the dual problem (9.5.4)–(9.5.5) is unbounded. Further, we know by the duality theory that the original problem (9.5.1)–(9.5.3) has no feasible point.

Now, by use of the analysis above, we describe the dual method due to Goldfarb and Idnani [155] as follows (we consider the case in which $m_e = 0$, i.e., the problem with only inequality constraints).

Algorithm 9.5.1 *(Dual Method)*

Step 1. $x_1 = -G^{-1}g, f_1 = \frac{1}{2}g^T x_1, \mathcal{S}_1 = \Phi; k := 1, \bar{\lambda}_1 = \Phi, q = 0.$

Step 2. *Compute* $r_i = b_i - a_i^T x_k,\ i = 1, \cdots, m.$
If $r_i \leq 0$, *stop.*
Choose p *such that* $r_p = \max_{1\leq i\leq m} r_i$;
$\bar{\lambda}_k := \begin{pmatrix} \bar{\lambda}_k \\ 0 \end{pmatrix}.$

Step 3. $d_k := \hat{G}_k a_p = G^{-1}(I - A_k A_k^*) a_p;\ y_k := A_k^* a_p.$
If $\{j|\,(y_k)_j > 0, j \in \mathcal{S}_k\}$ *is nonempty, set*

$$\alpha_k = \min_{\substack{(y_k)_j>0 \\ j\in\mathcal{S}_k}} \frac{(\bar{\lambda}_k)_j}{(y_k)_j} = \frac{(\bar{\lambda}_k)_l}{(y_k)_l}; \tag{9.5.28}$$

else set $\alpha_k = \infty$.

Step 4. *If* $d_k \neq 0$, *go to Step 5;*
If $\alpha_k = \infty$, *stop (the original problem has no feasible point);*
$\mathcal{S}_k := \mathcal{S}_k \setminus \{l\};\ q := q - 1;$
$\bar{\lambda}_k := \bar{\lambda}_k + \alpha_k \begin{pmatrix} -y_k \\ 1 \end{pmatrix};$
Modify A_k^* *and* $\hat{G}_k$; *turn to Step 3.*

Step 5 $\hat{\alpha} := -(b_p - a_p^T x_k)/a_p^T d_k;$
$\alpha_k := \min\{\alpha_k, \hat{\alpha}\};$
$x_{k+1} := x_k + \alpha_k d_k;$
$f_{k+1} := f_k + \alpha_k a_p^T d_k(\frac{1}{2}\alpha_k + (\bar{\lambda}_k)_{q+1});$
$\bar{\lambda}_{k+1} := \bar{\lambda}_k + \alpha_k \begin{pmatrix} -y_k \\ 1 \end{pmatrix}.$

Step 6. *If* $\alpha_k < \hat{\alpha}$, *go to Step 7;*
$\mathcal{S}_{k+1} := \mathcal{S}_k \cup \{p\}$; $q := q+1$;
Compute $\hat{G}_{k+1}$ *and* A^*_{k+1}, $k := k+1$; *turn to Step 2.*

Step 7. $\mathcal{S}_k := \mathcal{S}_k \setminus \{l\}$; $q := q-1$;
Remove the l*-th component from* $\bar{\lambda}_k$ *and obtain a new* $\bar{\lambda}_k$;
Compute $\hat{G}_k$ *and* A^*_k, *turn to Step 3.* $\square$

Next, we give a simple example which employs Algorithm 9.5.1.

Example 9.5.2

$$\min \quad \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}x_3^2 - 3x_2 - x_3 \tag{9.5.29}$$

$$\text{s.t.} \quad -x_1 - x_2 - x_3 \geq -1, \tag{9.5.30}$$

$$x_3 - x_2 \geq -1. \tag{9.5.31}$$

Solution. This example is a modification of the problem (9.3.16)–(9.3.18). The unique solution is still $(-1, \frac{3}{2}, \frac{1}{2})^T$. By use of Algorithm 9.5.1, we have

$$x_1 = -G^{-1}g = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix},$$

$$r_1 = 3 > 0, \ r_2 = 1 > 0.$$

Then we have $p = 1$ from Step 2, and

$$d_1 = G^{-1}a_p = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}.$$

Since $\mathcal{S}_1$ is empty, $\alpha_1 = \infty$ in Step 3. In Step 5, we get

$$\hat{\alpha} = -r_1/a_p^T d_1 = 1,$$

and obtain

$$\alpha_1 = 1, \ x_2 = x_1 + \alpha_1 d_k = \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix},$$

$$\bar{\lambda}_2 = (1), \ \mathcal{S}_2 = \{1\}.$$

Thus, after one iteration, x_2 is the solution of the subproblem

$$\begin{aligned} \min \quad & \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 + \frac{1}{2}x_3^2 - 3x_2 - x_3 && (9.5.32) \\ \text{s.t.} \quad & -x_1 - x_2 - x_3 = -1. && (9.5.33) \end{aligned}$$

In the second iteration, we have

$$r_1 = 0, \ r_2 = 1.$$

Then $p = 2$ from Step 2, and

$$d_2 = G^{-1}(I - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \frac{1}{3} (1\ 1\ 1)) \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

Since $y_2 = a_2^T a_1 = 0$, we have $\alpha_2 = \infty$ in Step 3. In Step 5, we have

$$\hat{\alpha} = -r_2 / a_2^T d_2 = \frac{1}{2}.$$

Then $\alpha_2 := \hat{\alpha} = \frac{1}{2}$,

$$x_3 = x_2 + \alpha_2 d_2 = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -\frac{3}{2} \\ \frac{1}{2} \end{pmatrix},$$

and $\bar{\lambda}_3 = (1\ \frac{1}{2})^T$. Hence, x_3 is the solution of the original problem and $\bar{\lambda}_3$ is the corresponding Lagrange multiplier. $\Box$

In concrete computation, Goldfarb and Idnani suggested using the Cholesky factorization of G,

$$G = LL^T,$$

and then employing QR decomposition to $L^{-1}A_k$, that is,

$$L^{-1}A_k = Q_k \begin{bmatrix} R_k \\ 0 \end{bmatrix}.$$

This approach allows us to get better numerical stability than by using G^{-1} directly.

Instead, Powell [274] suggested using

$$A_k = Q_k \begin{bmatrix} R_k \\ 0 \end{bmatrix} = [Q_k^{(1)}\, Q_k^{(2)}] \begin{bmatrix} R_k \\ 0 \end{bmatrix}, \tag{9.5.34}$$

and then employing the inverse Cholesky factorization of $[Q_k^{(2)}]^T G Q_k^{(2)}$, i.e.,

$$U_k U_k^T = [Q_k^{(2)}]^T G Q_k^{(2)}, \tag{9.5.35}$$

where U_k is an upper triangular matrix. In the algorithm that Powell [274] presented, each iteration updates $Q_k^{(1)}$, R_k, and U_k.

9.6 Interior Ellipsoid Method

Karmarkar [184] introduced a new polynomial-time algorithm for solving linear programming problems that sparked enormous interest in the mathematical programming community. Karmarkar's algorithm generates a sequence of points in the interior of the feasible region while converging to the optimal solution. This algorithm is effective and competitive with the simplex method in terms of solution time for linear programming (LP).

Ye and Tse [364] present an extension of Karmarkar's LP algorithm for convex quadratic programming. We introduce this algorithm in brief. The interested readers can consult the original paper for details.

The original version of Karmarkar's algorithm solves a linear programming of the special form

$$\min \quad \hat{c}^T \hat{x} \tag{9.6.1}$$

$$\text{s.t.} \quad \hat{A}^T \hat{x} = 0, e^T \hat{x} = n+1, \hat{x} \geq 0, \tag{9.6.2}$$

where $\hat{c} \in R^{n+1}, \hat{x} \in R^{n+1}, \hat{A} \in R^{(n+1)\times(m+1)}, e = (1, \cdots, 1)^T \in R^{n+1}$. Now, we generalize the Karmarkar's algorithm to convex quadratic programming.

Consider convex quadratic programming problem

$$\min \quad g^T x + \frac{1}{2} x^T G x \triangleq q(x) \tag{9.6.3}$$

$$\text{s.t.} \quad A^T x = b, \tag{9.6.4}$$

$$x \geq 0, \tag{9.6.5}$$

where $A \in R^{n\times m}$. Let x_k be an interior point, i.e.,

$$A^T x_k = b, \tag{9.6.6}$$

$$x_k > 0. \tag{9.6.7}$$

Define

$$D_k = \mathrm{diag}(x_k) = \begin{bmatrix} (x_k)_1 & & 0 \\ & \ddots & \\ 0 & & (x_k)_n \end{bmatrix}. \tag{9.6.8}$$

Let the transformation $\hat{x} = T_k x \in R^{n+1}$ as follows:

$$\hat{x}_i = \frac{(n+1)(D_k^{-1}x)_i}{e^T D_k^{-1} x + 1},\ i = 1, \cdots, n; \tag{9.6.9}$$

$$\hat{x}_{n+1} = (n+1)/[e^T D_k^{-1} x + 1]. \tag{9.6.10}$$

Obviously, the inverse transformation $T_k^{-1} : R^{n+1} \to R^n$ is defined by

$$x = T_k^{-1}\hat{x} = \frac{D_k \hat{x}[n]}{\hat{x}_{n+1}}, \tag{9.6.11}$$

where $e = (1, \cdots, 1)^T \in R^{n+1}, \hat{x}[n] = (\hat{x}_1, \cdots, \hat{x}_n)^T$. Then, the problem (9.6.3)–(9.6.5) can be written as

$$\min_{\hat{x}\in R^{n+1}} \quad \hat{x}_{n+1} q(T_k^{-1}\hat{x}) \triangleq \hat{q}(\hat{x}) \tag{9.6.12}$$

$$\text{s.t.} \quad A^T D_k \hat{x}[n] - \hat{x}_{n+1} b = 0, \tag{9.6.13}$$

$$e^T \hat{x} = n + 1, \tag{9.6.14}$$

$$\hat{x}[n] \geq 0,\ \hat{x} > 0. \tag{9.6.15}$$

By substituting (9.6.11) into (9.6.12), we obtain an equivalent form of (9.6.12)–(9.6.15):

$$\min \quad \hat{g}_k^T \hat{x}[n] + \frac{1}{2}\hat{x}[n]^T \hat{G}_k \hat{x}[n]/\hat{x}_{n+1} \tag{9.6.16}$$

$$\text{s.t.} \quad \hat{A}_k^T \hat{x} = \hat{b}, \tag{9.6.17}$$

$$\hat{x}[n] \geq 0,\ \hat{x}_{n+1} > 0, \tag{9.6.18}$$

where

$$\hat{G}_k = D_k G D_k,\ \hat{g}_k = D_k g, \tag{9.6.19}$$

$$\hat{A}_k = \begin{bmatrix} D_k A & \\ & e \\ -b^T & \end{bmatrix}, \ \hat{b} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ n+1 \end{pmatrix}. \tag{9.6.20}$$

Using the interior ellipsoid method, we solve the following subproblem (9.6.22)–(9.6.24) over an interior ellipsoid centered at $\hat{x}_k$, instead of solving the subproblem (9.6.12)–(9.6.15). Note that, for a given iterate x_k, $\hat{x}_k = T_k x_k = e$ and $\hat{q}(\hat{x}_k) = \hat{q}(e) = q(x_k)$. So, the interior ellipsoid happens to be an interior sphere in the feasible area of problem (9.6.12)–(9.6.15). Thus, the condition (9.6.18) can be enhanced to

$$\|\hat{x} - e\|_2 \le \beta < 1. \tag{9.6.21}$$

Obviously, (9.6.18) will hold provided (9.6.21) holds. Hence, we consider the subproblem

$$\min \quad \hat{g}_k^T \hat{x}[n] + \frac{1}{2}\hat{x}[n] + \frac{1}{2}\hat{x}[n]^T \hat{G}_k \hat{x}[n]/\hat{x}_{n+1} \tag{9.6.22}$$

$$\text{s.t.} \quad \hat{A}_k^T \hat{x} = \hat{b}, \tag{9.6.23}$$

$$\|\hat{x} - e\|_2 \le \beta < 1, \tag{9.6.24}$$

where $\beta < 1$ is a constant independent of k.

By the Karush-Kuhn-Tucker Theorem, solving (9.6.22)–(9.6.24) is equivalent to solving the following system:

$$\hat{g}_k + \hat{x}_{n+1}^{-1} \hat{G}_k \hat{x}[n] = \hat{A}_k[n]\lambda + \mu(\hat{x}[n] - e[n]), \tag{9.6.25}$$

$$-\frac{1}{2}\frac{1}{\hat{x}_{n+1}^2}\hat{x}[n]^T \hat{G}_k \hat{x}[n] = (\hat{a}_{n+1}^{(k)})^T \lambda + \mu(\hat{x}_{n+1} - 1) = 0, \tag{9.6.26}$$

$$\hat{A}_k^T \hat{x} = \hat{b}, \tag{9.6.27}$$

$$\|\hat{x} - e\|_2 \le \beta, \tag{9.6.28}$$

$$\mu[\|\hat{x} - e\|_2 - \beta] = 0, \ \mu \le 0, \tag{9.6.29}$$

where (9.6.25) and (9.6.26) are the first n equations and the last equation of the stationary point condition in KKT conditions respectively. Here $\hat{A}_k[n]$ is the matrix of the first n rows of matrix $\hat{A}_k$, $\hat{a}_{n+1}^{(k)}$ is the $(n+1)$-th row of $\hat{A}_k$, $e[n] = (1, \cdots, 1)^T \in R^n$ and $\lambda \in R^{m+1}$. So, (9.6.25) and (9.6.27) can be written in the following form

$$P_k \begin{bmatrix} \hat{x}[n] \\ \hat{\lambda} \end{bmatrix} = \hat{x}_{n+1}\bar{b} + \tilde{b}, \tag{9.6.30}$$

where

$$P_k = \begin{bmatrix} \hat{G}_k + \hat{\mu} I & -\hat{A}[n] \\ \hat{A}[n]^T & 0 \end{bmatrix}, \tag{9.6.31}$$

$$\bar{b} = \begin{bmatrix} -\hat{g}_k \\ b \\ -1 \end{bmatrix}, \ \tilde{b} = \begin{bmatrix} \hat{\mu} e \\ 0 \\ n+1 \end{bmatrix}, \tag{9.6.32}$$

$$\hat{\lambda} = \hat{x}_{n+1}\lambda, \ \hat{\mu} = -\hat{x}_{n+1}\mu. \tag{9.6.33}$$

Then, for any given $\hat{\mu} \geq 0$, we can find $\hat{\lambda}$ and $\hat{x}[n]$ from (9.6.30). Then we obtain $\hat{x}_{n+1}$ from substituting $\hat{\lambda}$ and $\hat{x}[n]$ into (9.6.26). This indicates that for any given $\hat{\mu} \geq 0$, we can find $\hat{x}(\hat{\mu})$. Define the function

$$h(\hat{\mu}) = \|\hat{x}(\hat{\mu}) - e\|_2 - \beta. \tag{9.6.34}$$

If $h(0) \leq 0$, then $\hat{x}(0)$ is the solution of (9.6.12)–(9.6.15). In this case, $x = D_k\hat{x}(0)[n]/\hat{x}(0)_{n+1}$ is the solution of the original problem.

If $h(0) > 0$, since $\lim_{\hat{\mu}\to\infty} h(\hat{\mu}) = -\beta < 0$, we can find $\hat{\mu}_k$ by a bisectioning method such that $h(\hat{\mu}_k) = 0$, and further the solution $\hat{x}(\hat{\mu}_k)$ of problem (9.6.22)–(9.6.24). By back-substituting $\hat{x}(\hat{\mu}_k)$, we obtain the new iterate x_{k+1}, i.e.,

$$x_{k+1} = T_k^{-1}\hat{x}(\hat{\mu}_k) = \frac{D_k\hat{x}(\hat{\mu}_k)[n]}{\hat{x}(\hat{\mu}_k)_{n+1}}, \tag{9.6.35}$$

where $\hat{x}(\hat{\mu}_k)[n] = (\hat{x}(\hat{\mu}_k)_1, \cdots, \hat{x}(\hat{\mu}_k)_n)^T$.

The interior ellipsoid algorithm for solving convex quadratic programming problems is introduced as follows.

Algorithm 9.6.1 *(Interior Ellipsoid Method for Convex QP)*

Step 1. Given a strict interior point x_1 of (9.6.3)–(9.6.5); $k := 1$.

Step 2. Solve the subproblem (9.6.22)–(9.6.24) for $\hat{x}(\hat{\mu}_k)$; and compute x_{k+1} by (9.6.35).

Step 3. If x_{k+1} is a KKT point, stop;
$k := k + 1$, go to Step 2. □

The further details of interior ellipsoid methods for convex quadratic programming can be found in Ye and Tse (1989).

9.7 Primal-Dual Interior-Point Methods

The primal-dual interior-point method for linear programming can be applied to convex quadratic programming through a simple extension of the method. Since we have not discussed the topic *linear programming* in the book, we first outline this method for linear programming.

Consider the linear programming problem in standard form

$$\begin{array}{rl} \min_{x\in R^n} & c^T x \\ \text{s.t.} & Ax = b, \\ & x \geq 0, \end{array} \tag{9.7.1}$$

where c and x are vectors in R^n, b is a vector in R^m, and A is an $m \times n$ matrix. The dual problem for (9.7.1) is

$$\begin{array}{rl} \max_{\lambda\in R^m} & b^T \lambda \\ \text{s.t.} & A^T\lambda + s = c, \\ & s \geq 0, \end{array} \tag{9.7.2}$$

where λ is a vector in R^m and s is a vector in R^n. The primal-dual solution of (9.7.1) and (9.7.2) are characterized by the Karush-Kuhn-Tucker (KKT) conditions:

$$A^T\lambda + s = c, \tag{9.7.3}$$

$$Ax = b, \tag{9.7.4}$$

$$x_i s_i = 0, \quad i = 1, \cdots, n \tag{9.7.5}$$

$$(x, s) \geq 0, \tag{9.7.6}$$

where vectors λ and s are Lagrange multipliers for the constraints $Ax = b$ and $x \geq 0$, respectively.

Primal-dual interior-point methods find primal-dual solutions (x^*, λ^*, s^*) of KKT system by applying variants of Newton's method to the three equality conditions (9.7.3)–(9.7.5) of this system and modifying the search directions and steplength so that the inequalities $(x, s) \geq 0$ are satisfied strictly at every iteration.

To derive primal-dual interior-point methods, we restate the KKT conditions (9.7.3)–(9.7.6) in a slightly different form by means of a mapping

$F : R^{2n+m} \to R^{2n+m}$:

$$F(x, \lambda, s) = \begin{bmatrix} A^T\lambda + s - c \\ Ax - b \\ XSe \end{bmatrix} = 0, \tag{9.7.7}$$

$$(x, s) \geq 0, \tag{9.7.8}$$

where

$$X = \mathrm{diag}(x_1, x_2, \cdots, x_n), \quad S = \mathrm{diag}(s_1, s_2, \cdots, s_n),$$

and $e = (1, 1, \cdots, 1)^T$. Note that F is actually linear in its first two terms $Ax - b, A^T\lambda + s - c$, and only mildly nonlinear in the remaining term XSe.

Primal-dual interior-point methods generate iterates (x^k, λ^k, s^k) that satisfy the bound (9.7.8) strictly, that is, $x^k > 0$ and $s^k > 0$. This property is the origin of the term *interior-point*. By respecting these bounds, the methods avoid spurious solutions, which are points that satisfy $F(x, \lambda, s) = 0$ but not $(x, s) \geq 0$.

Newton's method forms a linear model of F around the current point and obtains the search direction $(\Delta x, \Delta\lambda, \Delta s)$ by solving the following system of linear equations:

$$J(x, \lambda, s) \begin{bmatrix} \Delta x \\ \Delta\lambda \\ \Delta s \end{bmatrix} = -F(x, \lambda, s), \tag{9.7.9}$$

where J is the Jacobian of F. If the current point is strictly feasible, the Newton step equations become

$$\begin{bmatrix} 0 & A^T & I \\ A & 0 & 0 \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta\lambda \\ \Delta s \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -XSe \end{bmatrix}. \tag{9.7.10}$$

Note that a full step along this direction usually is not permissible, since it would violate the bound $(x, s) \geq 0$. To avoid this difficulty, we perform a line search along the Newton direction so that the new iterate is

$$(x, \lambda, s) + \alpha(\Delta x, \Delta\lambda, \Delta s) \tag{9.7.11}$$

for some line search parameter $\alpha \in (0, 1]$. Unfortunately, we often can take only a small step along the direction ($\alpha \ll 1$) before violating the condition $(x, s) > 0$. Hence the pure Newton direction (9.7.10) often does not allow us to make much progress toward a solution.

In the following, we give the central path technique which modifies the basic Newton procedure.

The Central Path

The central path $\mathcal{C}$ is an arc of strictly feasible points that plays a vital role in primal-dual algorithms. It is parametrized by a scalar $\tau > 0$, and each point $(x_\tau, \lambda_\tau, s_\tau) \in \mathcal{C}$ solves the following system:

$$A^T\lambda + s = c, \tag{9.7.12}$$

$$Ax = b, \tag{9.7.13}$$

$$x_i s_i = \tau, \quad i = 1, 2, \cdots, n, \tag{9.7.14}$$

$$(x, s) > 0. \tag{9.7.15}$$

These conditions differ from KKT conditions only in the term τ on the right-hand side of (9.7.14). Instead of the complementarity condition (9.7.5), we require that the pairwise product $x_i s_i$ have the same value τ for all indices i. From (9.7.12)–(9.7.15), we can define the central path as

$$\mathcal{C} = \{(x_\tau, \lambda_\tau, s_\tau) \mid \tau > 0\}.$$

It can be shown that $(x_\tau, \lambda_\tau, s_\tau)$ is defined uniquely for each $\tau > 0$ if and only if the strictly feasible set $\mathcal{F}^o$ defined by

$$\mathcal{F}^o = \{(x, \lambda, s) \mid Ax = b, A^T\lambda + s = c, (x, s) > 0\}$$

is nonempty. Hence the entire path $\mathcal{C}$ is well defined.

Another way of defining $\mathcal{C}$ is to use the mapping F defined in (9.7.7) and write

$$F(x_\tau, \lambda_\tau, s_\tau) = \begin{bmatrix} 0 \\ 0 \\ \tau e \end{bmatrix}, \quad (x_\tau, s_\tau) > 0. \tag{9.7.16}$$

The equations (9.7.12)–(9.7.15) approximate (9.7.3)–(9.7.6) more and more closely as τ goes to zero. If $\mathcal{C}$ converges to anything as $\tau \downarrow 0$, it must converge to a primal-dual solution of the linear program. The central path thus guides us to a solution along a route that steers clear of spurious solutions by keeping all the pairwise products $x_i s_i$ strictly positive and decreasing them to zero at the same rate.

Primal-dual interior-point algorithms take Newton steps toward points on $\mathcal{C}$ for which $\tau > 0$, rather than pure Newton steps for F. Since these steps are biased toward the interior of the nonnegative orthant defined by $(x, s) \geq 0$, it usually is possible to take longer steps along them than along the pure Newton steps for F before violating the positivity condition. To describe the biased search direction, we introduce a centering parameter $\sigma \in [0, 1]$ and a duality measure μ defined by

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i s_i = \frac{x^T s}{n}, \tag{9.7.17}$$

which measures the average value of the pairwise product $x_i s_i$. By writing $\tau = \sigma\mu$ and applying Newton's method to the system (9.7.16), we obtain

$$\begin{bmatrix} 0 & A^T & I \\ A & 0 & 0 \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta\lambda \\ \Delta s \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -XSe + \sigma\mu e \end{bmatrix}. \tag{9.7.18}$$

The step $(\Delta x, \Delta\lambda, \Delta s)$ is a Newton step toward the point $(x_{\sigma\mu}, \lambda_{\sigma\mu}, s_{\sigma\mu}) \in \mathcal{C}$, at which the pairwise product $x_i s_i$ are all equal to $\sigma\mu$. In contrast, the step (9.7.10) aims directly for the point at which the KKT conditions (9.7.3)–(9.7.6) are satisfied.

If $\sigma = 1$, the equations (9.7.18) define a *centering direction*, a Newton step toward the point $(x_\mu, \lambda_\mu, s_\mu) \in \mathcal{C}$. If $\sigma = 0$, the (9.7.18) gives the standard Newton step.

In the following, we define a general framework of primal-dual interior-point algorithm.

Algorithm 9.7.1 *(A Primal-Dual Interior-Point Framework)*

Given $(x^0, \lambda^0, s^0) \in \mathcal{F}^o$.

For $k = 0, 1, 2, \cdots,$

solve

$$\begin{bmatrix} 0 & A^T & I \\ A & 0 & 0 \\ S^k & 0 & X^k \end{bmatrix} \begin{bmatrix} \Delta x^k \\ \Delta\lambda^k \\ \Delta s^k \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -X^k S^k e + \sigma_k \mu_k e \end{bmatrix},$$

where $\sigma_k \in [0, 1]$ *and* $\mu_k = (x^k)^T s^k / n$;

set

$$(x^{k+1}, \lambda^{k+1}, s^{k+1}) = (x^k, \lambda^k, s^k) + \alpha_k(\Delta x^k, \Delta\lambda^k, \Delta s^k),$$

choosing α_k *such that* $(x^{k+1}, s^{k+1}) > 0$.

end(For). $\square$

For most problems, however, a strictly feasible starting point (x^0, λ^0, s^0) is difficult to find. Infeasible interior-point methods require only that the components of x^0 and s^0 be strictly positive. Therefore, we give a slight change to the equation (9.7.18). If we define the residuals for the two linear equations as

$$r_b = Ax - b, \quad r_c = A^T\lambda + s - c,$$

the modified step equation is

$$\begin{bmatrix} 0 & A^T & I \\ A & 0 & 0 \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta\lambda \\ \Delta s \end{bmatrix} = \begin{bmatrix} -r_c \\ -r_b \\ -XSe + \sigma\mu e \end{bmatrix}. \tag{9.7.19}$$

Primal-Dual Interior-Point Methods for Convex Quadratic Programming

Now we return to convex quadratic programming. Let us discuss convex quadratic programming with inequality constraints:

$$\min_{x\in R^n} \quad q(x) \stackrel{\text{def}}{=} \frac{1}{2}x^TGx + x^Tg \tag{9.7.20}$$

$$\text{s.t.} \qquad Ax \geq b, \tag{9.7.21}$$

where $g \in R^n, b \in R^m, A \in R^{m\times n}$ and $G \in R^{n\times n}$ is symmetric and positive semidefinite. The KKT conditions of (9.7.20)–(9.7.21) state as follows: If x^* is a solution of (9.7.20)–(9.7.21), there is a Lagrange multiplier vector λ^* such that the following conditions are satisfied for $(x, \lambda) = (x^*, \lambda^*)$:

$$Gx - A^T\lambda + g = 0, \tag{9.7.22}$$

$$Ax - b \geq 0, \tag{9.7.23}$$

$$(Ax - b)_i\lambda_i = 0, \quad i = 1, 2, \cdots, m, \tag{9.7.24}$$

$$\lambda \geq 0. \tag{9.7.25}$$

By introducing the slack vector $y = Ax - b$, we have

$$Gx - A^T\lambda + g = 0, \tag{9.7.26}$$

$$Ax - y - b = 0, \tag{9.7.27}$$

$$y_i\lambda_i = 0, \quad i = 1, 2, \cdots, m, \tag{9.7.28}$$

$$(y, \lambda) \geq 0. \tag{9.7.29}$$

As in the case of linear programming, here the KKT conditions are not only necessary but also sufficient, because problem (9.7.20)–(9.7.21) is convex programming. Hence we can solve it by finding solutions of system (9.7.26)–(9.7.29). As discussed above, we apply modifications of Newton's method to this system. We can define

$$F(x, y, \lambda) = \begin{bmatrix} Gx - A^T\lambda + g \\ Ax - y - b \\ Y\Lambda e \end{bmatrix}, \quad (y, \lambda) \geq 0, \tag{9.7.30}$$

where

$$Y = \operatorname{diag}(y_1, y_2, \cdots, y_m), \quad \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \cdots, \lambda_m), \quad e = (1, 1, \cdots, 1)^T.$$

Given a current iterate (x, y, λ) that satisfies $(y, \lambda) > 0$, we can define a duality measure μ by

$$\mu = \frac{1}{m}\sum_{i=1}^{m} y_i\lambda_i = \frac{y^T\lambda}{m}. \tag{9.7.31}$$

The central path $\mathcal{C}$ is the set of points $(x_\tau, y_\tau, \lambda_\tau)(\tau > 0)$ satisfying

$$F(x_\tau, y_\tau, \lambda_\tau) = \begin{bmatrix} 0 \\ 0 \\ \tau e \end{bmatrix}, \quad (y_\tau, \lambda_\tau) > 0. \tag{9.7.32}$$

The generic step $(\Delta x, \Delta y, \Delta\lambda)$ is a Newton-type step toward the point $(x_{\sigma\mu}, y_{\sigma\mu}, \lambda_{\sigma\mu}) \in \mathcal{C}$. As in (9.7.19), this step satisfies the following system:

$$\begin{bmatrix} G & -A^T & 0 \\ A & 0 & -I \\ 0 & Y & \Lambda \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta\lambda \end{bmatrix} = \begin{bmatrix} -r_g \\ -r_b \\ -\Lambda S e + \sigma\mu e \end{bmatrix}, \tag{9.7.33}$$

where

$$r_g = Gx - A^T\lambda + g, \quad r_b = Ax - y - b.$$

So, we obtain the next iterate

$$(x^+, y^+, \lambda^+) = (x, y, \lambda) + \alpha(\Delta x, \Delta y, \Delta\lambda), \tag{9.7.34}$$

where α is chosen so that $(y^+, \lambda^+) > 0$.

For mode details of primal-dual interior-point methods for convex quadratic programming, please consult Wright [358].

Exercises

1. Let $H = \mathrm{Diag}(h_{11}, h_{22}, ..., h_{nn})$ be a positive definite diagonal matrix. Find the minimizer of (9.1.1) subject to the condition $\|x\|_\infty \le 1$.

2. Prove Theorem 9.1.1.

3. Prove that problem (9.2.8)–(9.2.10) is the dual of problem (9.1.1)–(9.1.3).

4. Solve the dual of the problem

$$\begin{array}{ll} \min & (x_1^2 + x_2^2)/2 + x_1 \\ \text{s.t.} & x_1 \ge 0. \end{array}$$

5. If $f(x)$ is a convex function and $c_i(x) (i = 1, \cdots, m)$ are concave functions, the problem

$$\begin{array}{ll} \min & f(x) \\ \text{s.t.} & c_i(x) \ge 0 \end{array}$$

is called a convex programming problem. Generalize the dual theory in Section 9.2 from convex quadratic programming to general convex programming.

6. Find the smallest circle in the plane that contains the points $(1, -4)$, $(-2, -2)$, $(-4, 1)$ and $(4, 5)$. Formulate the problem as a convex programming problem, then solve the dual.

7. Assume that $B \in R^{n\times n}$ is positive definite, $A \in R^{m\times n}$, $g \in R^n$ and $b \in R^m$. Give the dual problem of the following QP:

$$\begin{array}{ll} \min & g^T x + \frac{1}{2} x^T B x \\ \text{s.t.} & Ax = b. \end{array}$$

8. Solve the equality constraint QP problem

$$\begin{array}{ll} \min & \begin{pmatrix} 1 \\ -1 \end{pmatrix}^T x + \frac{1}{2} x^T \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} x \\ \text{s.t.} & x_1 + x_2 = 1. \end{array}$$

9. 1) Check that if set $V = \begin{bmatrix} 0 \\ I_{n-m} \end{bmatrix}$ in (9.3.42), the choice (9.3.37)–(9.3.38) can be obtained.

2) Check that if set $V = Q_2$ in (9.3.42), the choice (9.3.40) can be obtained.

10. Show that if $A \in R^{n\times m}$ has full column-rank and Z^TGZ is positive definite, then KKT matrix (9.3.47) is nonsingular.

11. Show (9.3.53)–(9.3.55).

12. Show (9.3.58)–(9.3.60).

13. Assume $A \in R^{m\times n}$ has full row rank. Let $Z \in R^{n\times(n-m)}$ be any full column rank matrix such that $AZ = 0$. Prove that the matrix

$$\begin{bmatrix} B & A^T \\ A & 0 \end{bmatrix} \tag{9.7.35}$$

is nonsingular if and only if Z^TBZ is nonsingular.

14. Use the active set method to solve the problem

$$\begin{aligned} \min \quad & -1000x_1 - 1000x_2 + x_1^2 + x_2^2 \\ \text{s.t.} \quad & 3x_1 + x_2 \geq 3, \\ & x_1 + 4x_2 \geq 4, \\ & x_1 \geq 0, \\ & x_2 \geq 0. \end{aligned}$$

Illustrate the result by sketching the set of feasible solutions.

15. Program Algorithm 9.4.2 and use it to solve

$$\begin{aligned} \min \quad & x_1^2 + 2x_2^2 - 2x_1 - 6x_2 - 2x_1x_2 \\ \text{s.t.} \quad & \frac{1}{2}x_1 + \frac{1}{2}x_2 \leq 1, \\ & -x_1 + x_2 \leq 2, \end{aligned}$$

$$x_1, x_2 \geq 0.$$

16. Try to give the primal-dual interior-point algorithm for convex quadratic programming (9.7.20)–(9.7.21).